

EBNAT-KAPPEL, Switzerland

WMO: 066930

Lat: 47.2733N

Lon: 9.1086E

Elev: 2044

StdP: 13.64

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.5	13.8	4.3	7.4	11.9	10.3	10.0	17.1	11.6	39.3	9.9	34.3	2.9	150	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	85.3	68.3	81.9	66.8	78.6	65.7	69.9	82.5	68.1	79.6	66.5	77.0	4.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	101.9	75.8	63.9	96.1	73.2	62.6	91.6	71.2	35.2	82.8	33.6	79.7	32.3	77.5	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.5	1.1	89.8	6.3	2.0	-3.4	91.3	-7.1	92.5	-10.6	93.6	-15.2	95.1		
(5)			0.7	73.1	6.2	1.7	-3.8	74.4	-7.4	75.4	-10.9	76.4	-15.4	77.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	31.5	31.1	39.8	47.2	53.2	61.2	64.0	63.8	56.2	48.9	39.8	33.8		
	DBStd	13.42	7.59	8.89	6.02	6.49	6.15	5.95	5.55	5.45	5.17	6.42	6.83	6.40		
	HDD50	2501	574	528	316	127	38	1	0	0	7	97	311	502		
	HDD65	6477	1039	948	780	534	369	144	84	90	266	499	757	967		
	CDD50	1639	0	0	1	43	135	337	434	428	193	63	5	0		
	CDD65	139	0	0	0	0	2	30	53	53	1	0	0	0		
	CDH74	1951	0	0	0	11	67	488	698	628	58	1	0	0		
	CDH80	503	0	0	0	0	5	135	199	164	0	0	0	0		
(14) Wind	WSAvg	2.4	2.4	2.4	2.8	2.9	2.7	2.7	2.4	2.2	2.1	2.1	2.1	2.1		
(15) Precipitation	PrecAvg	60.2	3.6	3.8	4.3	4.3	6.3	6.8	6.8	6.5	5.0	4.1	4.3	4.4		
	PrecMax	77.6	8.2	10.7	10.1	8.3	13.8	10.3	12.7	11.2	9.4	8.1	9.4	10.6		
	PrecMin	47.5	0.3	1.1	1.5	0.7	1.7	2.8	2.6	1.1	2.3	0.3	0.0	0.2		
	PrecStd	6.9	2.2	2.1	2.0	2.0	2.8	2.0	2.6	2.6	1.8	1.9	2.2	2.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	56.1	65.0	75.0	80.3	88.7	90.1	88.0	78.3	71.2	64.4	56.4		
		MCWB	47.0	44.6	50.3	57.2	62.3	70.2	69.8	68.5	65.7	56.7	58.1	43.5		
	2%	DB	46.5	50.2	60.8	70.4	75.1	83.9	85.2	84.4	75.7	67.3	59.5	49.6		
		MCWB	42.6	42.6	48.5	54.9	60.3	67.8	68.1	68.5	64.1	57.3	51.4	44.0		
	5%	DB	43.1	45.7	57.1	67.5	71.1	79.8	81.4	81.0	71.8	63.8	54.4	46.3		
		MCWB	40.7	40.2	47.0	53.9	57.8	66.0	66.6	67.5	62.3	55.9	49.4	42.8		
	10%	DB	40.5	41.8	52.5	63.0	67.8	75.4	77.3	77.0	67.8	59.3	49.9	43.0		
		MCWB	38.9	38.1	44.9	51.9	56.2	63.7	64.7	66.5	60.4	54.5	46.9	40.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.0	48.3	52.0	58.5	64.5	72.8	71.6	71.3	67.0	60.7	55.9	47.6		
		MCDB	51.8	53.5	61.0	70.9	77.0	85.8	85.3	82.7	76.5	67.1	63.9	52.5		
	2%	WB	43.1	43.6	49.7	56.3	61.6	69.0	69.5	69.7	64.7	58.3	51.8	45.4		
		MCDB	45.6	48.5	57.7	67.8	72.4	80.8	83.0	81.7	73.2	64.2	57.0	48.8		
	5%	WB	41.1	41.0	47.7	54.5	59.3	66.6	67.6	68.2	63.1	56.7	49.5	42.9		
		MCDB	42.8	45.5	55.2	65.5	68.6	78.3	78.9	79.3	70.9	61.9	54.0	46.4		
	10%	WB	39.1	38.5	45.8	52.4	57.3	64.5	65.7	66.6	61.0	54.9	47.3	40.8		
		MCDB	40.5	41.2	52.3	62.2	66.0	74.1	75.7	76.6	67.1	59.0	50.1	43.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	14.4	19.2	21.5	20.2	20.3	20.5	20.9	19.3	17.5	14.0	13.7		
		MCDBR	14.3	21.5	29.2	31.8	29.6	29.9	28.4	28.4	25.9	25.3	22.3	18.1		
		MCWBR	10.9	13.2	17.6	17.5	15.8	14.8	13.0	14.0	15.3	15.4	14.7	12.9		
	5% WB	MCDBR	13.5	19.1	26.4	29.9	26.0	27.4	26.1	26.5	24.2	22.4	20.5	16.9		
		MCWBR	10.7	12.6	16.6	17.5	14.5	14.1	13.0	13.7	14.8	14.6	14.0	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.304	0.342	0.381	0.396	0.403	0.405	0.397	0.367	0.338	0.306	0.275		
	taud		2.401	2.339	2.316	2.278	2.286	2.308	2.299	2.339	2.402	2.468	2.487	2.429		
	Ebn at Noon		259	273	278	276	275	273	271	268	267	258	246	246		
	Edh at Noon		22	29	34	39	40	40	40	36	31	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		394	669	1053	1382	1570	1809	1760	1535	1159	737	420	325		
	RadStd		44	88	143	203	158	167	159	154	118	81	53	48		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (122 neighbors)	N/A	N/A	N/A	+1.26	+0.76	+0.59	N/A	N/A	+103	+27

Nomenclature: See separate page